

Active Inference Institute ~ 2024 Quarterly Roundtable #3

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hello and welcome it's October 1st

2024 and we're in quarterly Roundtable

number three for the active inference

Institute welcome to the ACM Institute

we're a participatory online Institute

that is communicating learning and

practicing applied active inference you

can find more information at these links

this is recorded in an archive live

stream

so please provide feedback so we can

improve our work all backgrounds and

perspectives are welcome and we'll

follow a video etiquette for live

stream okay today in the quarterly Round

Table we'll go over a few Institute

updates look at some developments

occurring in projects and then talk a

little bit about what'll be happening in

the last quarter of 2024 how we'll be

setting up for next year before we jump

in blue or Alex want to make any

comments or say

hello sure hi uh excited to round this

year out with the ecosystem paper and

the Symposium and um yeah looking

forward to 2025 and all the updates

we'll make

then yeah hey everyone and possibly I

will bring it into the live what we

shortly discuss in the beginning and

before the stream that uh looks like for

our current and next uh round tables we

will talk more and more about some

project

results and that's great that we see more and more projects and more developments not only some specific plans all right so people watching live feel free to write any questions or anything in the live chat we'll keep an eye on it

all right first at The Institute scale here's an overview slide you can screenshot a bunch of stable links that'll help start you on your journey email us join the Discord check out some projects learn about the ecosystem find out more structured ways to get involved stay updated with video or newsletter check out fellows programs organizational Partnerships and provide financial support bunch of short links

okay as blue alluded to our main Institute level focus and collaboration for the next six weeks or so is the versioning of this ecosystem and Institute paper so we wrote the first version as a big team in 2023 with visuals and a lot of different discussion and hammering it out a lot of really interesting points and now we're in the versioning the first versioning heading towards the second snapshot version of what'll be a large living document that'll provide many entry points and many paths through different things happening at the Institute and also at the ecosystem and clarify the relationships between them like what would be impactful for the active inference ecosystem to do what kinds of capacities could it build what kinds of

differences could it make what would the Institute have to do and be in order to support that and what would we need from a support or participation perspective for how that would come to be so over the next six weeks there'll be a lot of collaboration and discussion synchronous and asynchronous no one is going to add everything to every page but just like it always is except more this is a project where we just jump in and make many light touches and it's been awesome already to see how people have added things to these different pages and made structural suggestions there's one week left if you want to register go to ecosystem. active influence. Institute and fill out the form and you'll jump in and then for the last five weeks with whoever's on board then there's already 20 or 30 people who will be co-authors and are writing in these pages over the next 5 weeks we'll come together set up a few synchronous and asynchronous ways of people focusing their contributions and then we'll snapshot the version two in the days before the Symposium and it's really an important way for people whether they've been largely on the sidelines watching or whether they've been involved for years how can we hear from people at the end of this year and help set a course for next year so there'll be new visuals new fun discussions blue or Alex want to give any thoughts on this document or how we use it or or anything else about it uh you mentioned visuals and yes I

still see in this picture we have from a last uh

version uh it looks very nice in terms of visuals but also it was uh defining structure for our activities how we see Institute as organization which how it operates and how it's breakdowns in organizational sense which functions we need to parate such entity and so on everything was uh like basically in place and uh somehow planned for current vision and based on experience from the year working in projects and changes in ecosystem definitely we will work on contents of the paper but finally we will need to update visuals and I hope our new visuals will be even more attractive and defining in terms of structure for our future

works thank you

blue well that's a tough act to follow uh no I'm just excited to see what pans out and um you know it's it's nice for people to put their expectations for the institute for the next steps for where they think we are right now it's nice to see that like from other people bes besides just us uh it kind of lets us update like oh well maybe we're we are where we think we are or maybe we're not where where we think we are and what could we do to be better so it's a fun process also without going too much into it it's amazing how much the document writing environment has evolved in the last year like a year ago we were talking about a linear document

and how to make it readable and accessible all these different kinds of considerations then take that document already in the last few months I was able to smash it up against a bunch of Grant openings and get quite a ways towards translations into many languages translations to many audiences applications to Grants all these different sorts of things augmented to a pretty great first draft status using llms so having these kinds of comprehensive documents and living documentation that can have accurate information about what is the organization who's available who has what kinds of capacities that kind of information can flow directly in uh Grant Pathways so it's just really exciting how how that can play out all right second major piece of the coming few months and weeks will be the fourth applied active inference Symposium it's going to be November 3 1 to 15

2024 uh it'll be free and online we also hope that there'll be some satellite physical events for people who want to do a watch party or live stream from some physical location the uh opening session will be with Carl Friston there'll be first uh session on just recapping what is the application environment look like what has changed since 2021 when we had the first applied active Symposium with Carl uh what's happened this past year and then the second and third hour will be live

stream 58.2 so that'll be the third discussion that we will be having on the renormalization group advances and then there's going to be a whole host of other

presenters sharing everything from tools and their case studies their updates their successes and challenges in applying actim and also we'll have a lot of hackathon open source code collaboration

opportunities it will be very welcome and open for beginners and all questions and we'll look to curate questions and help make sense of things together and we'll be sharing more information on that in

October anything else that either of you'd like to add about the

[Music]

Symposium um that's great that we have uh like this initiative became stable and we have each year we have Symposium uh but if you remember we like initially trying to choose format which will be most valuable for our online community which is growing and so on but from very beginning we was focused on that uh applied uh stance of uh of the topics uh because we understand that which rapidly grow in field in terms of theory and researches researchers working in the field but finally from uh like systems engineering viewpoints we always need to have some understanding how it should be applied finally and this year I believe in terms of developments of tools and

Frameworks like Eric
siner uh it's really became more and
more close to real life applications and
we see around
some like prototypes and some artificial
Bots are working their on
offline and yes we need to be Aline uh
with ecosystem with our partners and
participants and so on
and see this have his focus on applying
active inference and che check like uh
each year with each others in a field to
share developments and
perspectives that's an exciting like
sneak preview about the renormalization
group too I'm excited to uh to peep
that yeah and it's just exciting to hold
the
open and plural space where we have
research developments and applications
all the way on through domain experts
and it's one thing to hear about well
active inference could be applied to
supply chain here and to this kind of
thing here and then having that two-way
street with a domain expert and maybe
just their verbal account of the
situation or the systems modeling
they've already done with the kinds of
ontologies narratives formal documents
tools Etc that they already have may
even be closer than many people suspect
to having a functioning active inference
model so it's a great time we'll touch
on some of these points uh coming up but
again if you'd like to be a presenter or
if you'd like to facilitate some other
kind of interactive session if you'd

like to be a sponsor we can share information on that just go to the Symposium page and that's where information will

be uh a quick note here on important developments of the format of Symposium itself I believe it's about that uh hakon sessions and we will put additional attentions and efforts to organize this part of Symposium in best possible way uh and it's kind of development of the format and really possibly in a few years uh the most uh activities of such symposiums will be on on hakon site

see yeah thanks I I agree live chats are awesome and the the live interaction of live streams is awesome also a lot of trainings can be done like model streams ahead of time and having symposia and convenings where people can just rapidly break through many many layers of uncertainty it can go from like listening to hours of podcasts or reading papers and thinking okay it could apply here but I don't really see how or I don't even even see that I don't see how and then all of a sudden when you're on the team and talking to people and the code is out and that the sketching is happening it could just happen so fast and it's just really exciting learning

experience okay about research fellows so our three current fellows Anna John FIS and John Boy are still active they've been research fellows for a couple of months now and our next round

of review is going to be in about 2 weeks so if you're considering putting in a research fellow application and you want to be considered for this second cohort get in contact whenever you can and uh we'll continue with a rolling deadlines around research fellows and other kinds of fellows so it's been awesome to see each of the three continue their their projects in very unique ways and there will be a lot more to come with more fellows on organizational

Partnerships we've continued and have a have a great collaboration with first principles first with John Clippinger and the FP1 project and several other discussions with organizational Partners these are with financial or within kind donations and support and there's a lot of opportunities for different organizations that want to contribute to the open source ecosystem want to learn and apply actim faster want to give strategic input to The Institute and access the community all these different sorts of things that Partners can help synergize and and um collaborate on any comments on anything

no uh let me add something that yes that two important Direction uh of research fellows and partnership with organization are strategic for this year and we see results see uh work working of people on our aligned goals uh that's great and we will develop this direction yes for research fellows it's two weeks

to make decisions and put your
application if you see some perspectives
and we as institution will support you
in this way
and for Partnerships with organization
yes we will uh continue this way and
always happy to see in new like players
in ecosystem and if you as
organizational entities see
some affordances which can be realized
together let's
discuss cool yeah a lot of things
happening with us writing letters of
support or being possible contractors on
grants range to us applying for Grants
and including other organizations so
doing together what we can't do
alone our philanthropy continues donate.
active inference do Institute now has
some nice drop-down options for
supporting a certain project with a
restricted donation so this means that
the funds for the donation go to the
project um we're a
501c3 organization that maybe matter for
some donors and while we are applying to
Grants some of which we'll describe
later on we're currently an all
volunteer operation and donations are
and will be vital to ensure that there's
a sustainability to what we're working
on so onward we go with philanthropy any
comments here
and I'm bringing in a live Super Chat
from
upcycle thank you
upcycle our philanthropy program at
work nice um some updates on our uh

social

media the project prepare and measure

form continues to operate like with the

fellows the partners several other

aspects that we've been

discussing I'm really happy we have

these quarterly round tables and the

newsletters it's easy to forget how

quickly these structural changes come

into play it was only this year when a

lot of those structures and affordances

came into being so we continue to see

many people propose projects some of

which we'll review and having this

always open measure opportunity for

people to share ranging from what's

happening in their internship to their

research lab to their company and these

channels will just continue to be

important as the ecosystem grows and

just like any other ecosystem no

organism is in all places at once so

having a culture of sharing what is

intended to be shared and having the

ways for people to select and using

synthetic methods to to expand and

distill as needed there's some extremely

exciting

possibilities just on our oh Alex SK

for uh yeah possibly get back to yeah to

this slide and I just take a place of

some newcomer which is first time on our

stream of such discussion and see seen

this picture and this structure and

connections uh to some specific

organizational process what we have it

could be not clear but for me it's a

great picture what we really have pieces

changes in our organizational process
which is based on our first principal
materials what we have in Institute and
having understanding and seeing now how
it's starting to work I just want to
highlight for people who possibly not
understand from the beginning what is
presented here it's really first
principles in works of organization and
possibly for your organization it's
could be interested to take a course of
physics as information processing and
take some principles uh into your work
as we done it for
hours thanks

here we have kind of a formal
organizational summary with the bodies
and The rhythms the scientific Advisory
Board the officers and the board of
directors we had a board of directors
meeting last week that was very
insightful and in the coming weeks and
months we'll be preparing and already
the applications and the information on
these positions exists on our website
and we'll be adding more information and
clarifying like what are these volunteer
positions what would be the kinds of
pragmatic and epistemic value that
somebody might glean or gain or
contribute from from working and
contributing this

way

okay those are the Institute type
activities now we'll have some shorter
project
oriented

sections so here are the Ed active

projects at The Institute we have the ontology project this has been continuing had many fun conversations in the Discord people learning using AI using natural methods to explore the ontology audiovisual production will come to textbook group will come to and active inference Journal also continuing to to explore what's possible with developing and augmenting transcripts in the textbook group we continue cohort six and cohort 7 continue on and as we're now about 2 and 1 half years out from the publication of the textbook and closing in on six and a half cohorts completed we're thinking a lot about how does this role that the textbook served in 2022 and 2023 how is that updating in 24 as people who have been through it one or three cohorts start to move on to other research projects what kinds of entry points exist and how how can we continue to to use the templates and the approaches like using generative AI to help unpack math as well as the the math learning group that octopus is facilitating how can we use the ontology to help connect natural language readings of equations with different kinds of situations and studies how can we Cate questions and discourse that people ask and and help promote collaboration all these different ways so the group's still active and also it'll be continuing to to develop as we change our uh offerings depending on

what people want to do and what
facilitators want to contribute
what in production we've had 26 videos
over the last few months for a total of
539 videos
overall here were the streams from the
last few
months there were 1 two 3 four five math
art streams with shanana Dobson and
Friends covering a a wide variety very
exciting and all over the place and math
already in the active inference streams
I've been working in the active inference
code repo developing open source code
for a a variety of different
settings we've had 11 guest
streams on topics like reinsurance that
was yesterday thrilling thrilling
reinsurance live
stream limits of the brain and
predicting events fellow John Bo with a
cognitive narrative
ecosystem guest stream 86 on biological
neurons and how biometric and and
heterogeneous Computing methods contrast
with reinforcement
learning coming more from a
computational side David Blumen and also
talking about reinforcement learning and
active inference Robert
Warden with a pair of live streams on
the the projective wave theory Raph and
Charles talking about different kinds of
cognition in in natural and ILM type
systems fellow Anna talking about her
ongoing Book Project and Josh Bongard
talking about embodied intelligence
really cool guest

streams and
then several other
streams ran way in a model stream giving
one of the most distilled and Powerful
unpacking of how active inference is
similar and different than reinforcement
learning and meta RL
methods uh discussion of quantum infer
ants and about Quantum measurements and
Quantum information studies and active
inference Toby Sinclair SMI from the
category theory side discussing
structured active inference which is
very promising way that active inference
methods are starting to be appli to apis
and other kinds of computational
settings uh the
closing season finale SL reflection with
Darius from insights turning the tables
asking him about how the insights series
went and and those uh 23 prior episodes
were some of the coolest and and most
viewed
conversational uh onboarding points for
a lot of different thinkers and ideas
and two art streams on on different work
at the intersection of predictive
processing active Aesthetics
art a lot of fun discussion so if
anyone's interested
in the planning part in inviting guests
or curating the suggestions from others
and just helping follow through on
invites and just getting things planned
if you're interested in audio visual
production manual craft production or
using synthetic methods to produce in
ways that people have not been able to

produce before if you want to help on the journal side there's a lot of ways to help improve this whole process okay on to Reiner updates here the Reiner projects uh these are projects where where people can get Hands-On with coding and with knowledge engineering it's been a productive month for the RX andf fur. JL learning group at the at the Institute this short URL will pop you into our Kota where there's always uh developments and things being posted but this month we submitted two grants with some partially overlapping teams that was exciting um earlier in September we submitted a farmworks decentralized AI agents for personalized solutions that was a future of Life Institute grant program about uh centralized Ai and power asymmetries and also pretty fun to then put up the grant as a pre-print so whether we get it or not we're co-authors we go on with a show and uh we can communicate that around without even knowing whether we've gotten it or not and we don't and then just a few days ago we submitted a uh multi-agent AI safety Grant to a foresight Institute program and this one that the tentative meme project name was Village validating inference for large scale agent governance ecosystems so we'll see where this goes and even apart from the grant applications there have been just many developments with RX and fur and the

application ecosystem so it's quite exciting to see where it's all going and then here's just the other projects that are in the actm ecosystem

uh really cool that people have proposed all these different things and and so much also happens outside of these tables so big thanks to all those who have put up their projects so that others can join in and learn more about them and uh put in the the measurements and that's how we'll share more specific detail any more comments blue or

Alex no I'm good

thanks okay for live Chatters feel free to write any questions otherwise here's just one final slide

here we are looking in the mirror Life's a blur even if we could record it all what would it really say and so all these different views and and angles that we take dayto day how is active inference helping us navigate this how can surprise pragmatic and epistemic value how can all all of these things that we're talking about be their own intrinsic meaningful contributions and also help us in the in the times between This Is A View From The topos Institute blue and I went to a small meeting there just a few weeks ago learning about category Theory but that's what it looks like from the nth floor of a building in Berkeley

and the walk continues through whatever
we suppose it to
be

um I'll wait a minute just if there's
any live chat questions or anything
people that uh want
to ask otherwise um what would be your
closing thoughts blue and

[Music]

Alex um possibly I will uh get back to
educ education and that experience of of
textbook

group and yes uh right questions you
raised about how we should use that
experience from that learning on
specific textbook but also I believe
that during that way we set up and
tested and evolved uh uh the methodic
process itself and currently we for our
needs if you'll see some new textbook or
course we can organized uh group
learning in the same way and possibly
from some moment it's also could be
affordance from some other projects or
institutes or some someone who who are
searching looking for new approach on
such collaborative ways of learning and
we can share possibly like templates in
Koda how to enabled or have some other
provide some other support for such
activities for any others valuable
knowledge

yes at like multiple scales I mean if
there's a time slot for something that
you want to jump in and stick by make
the suggestion if there's another
textbook group with a a similar
structure another book or another

another way it it's all whatever
volunteers step up to do so we have a
lot
of
latitude don't let the structures in
play give a blind spot to the degrees of
freedom that still
exist
blue for me I'm just excited like to see
um you know every time like we're
rounding out the year and we're coming
to like the close of our activities for
24 and um like where will we be a year
from now is always like like I'm always
surprised by how far we've come and not
only the Institute but overall um just
what what's possible and and um coming
to the Forefront with all the synthetic
knowledge uh and and what does that mean
and and like will synthetic knowledge be
um as useful more useful will people
like want real knowledge again someday
like a human human generated like will
We crave that will it will it become so
pervasive or I don't know I'm just
always curious to see um where where
that's taking us because it seems like
uh kind of the cat is out of the bag or
the Pandora's Box has been opened and um
the possibilities really seem seem
endless so I'm excited to see what where
we where we're at at this time in 25 and
um you know how how much will have been
done and how much will have changed and
and then the court that is still the
same
awesome okay
so stable links to check out if you're

watching in nearish real time join on in
to the ecosystem project or come along
for another versioning
run similarly for the
Symposium okay thanks and Alex thank you
upcycle everyone else okay bye thanks
everyone thank you